

HONGHAO ZHAO

+1 (458) 209-9738 | hhzhao46@gmail.com | [LinkedIn](#)

Education

Johns Hopkins University

Master of Science in Engineering in Computer Science

Aug 2024 - May 2026

GPA: 3.75/4.00

Boston University

Bachelor of Science; Double Major in

Pure and Applied Mathematics and Computer Science

Sep 2021 - Jun 2024

GPA: 3.70/4.00

Research Experience

Alliance AI Lab, Johns Hopkins University

Research Assistant | Mentor: Dr. Haimin Hu

Jun 2026 - Present

- Develop a cognitive layer for the Human-Centered Safety Filter (HCSF) by fine-tuning vision-language models and integrating haptic feedback with visual and voice cues to enable a Tri-level Human-Centered Safety Filter (Tri-HCSF).

NeuroAgent AI

Research Intern | Mentor: Dr. Kemar Green

May 2026 - Present

- Contribute to computational sensing and computer vision research supporting precision neurology and digital biomarker initiatives.

LCSR, Johns Hopkins University

Research Assistant | Mentors: Dr. Peter Kazanzides and Dr. Ugur Tumerdem

Oct 2025 - Present

- Co-developed a digital-twin VR teleoperation system for the dVRK, combining markerless perception and multi-view visualization for surgical robot control.
- Built AMBF simulation and data collection pipelines; extended ACT with force inputs and evaluated imitation learning for autonomous surgical manipulation.

Intuitive Computing Laboratory, Johns Hopkins University

Research Assistant | Mentor: Dr. Chien-Ming Huang

Apr 2025 - Sep 2026

- Developed data processing and clinician-facing analytics for LLM-powered sleep diaries through co-design with sleep specialists.
- Contributed to DREAM (Discovery, Reflection and Education for Awareness and Management of Sleep), a voice-based assistant supporting guided reflection and personalized sleep education.

PUBLICATIONS & Preprints

*Equal contribution.

1. Chang Liu*, Chenhao Yu*, **Honghao Zhao***, Hao Ding, Haochen Wei, Adnan Munawar, Mathias Unberath, Peter Kazanzides. **Digital Twin-Driven VR Teleoperation with Multi-View Spatial Perception for Surgical Robots.** *Research manuscript*, 2026.
2. Amama Mahmood, Bokyung Kim, **Honghao Zhao**, Molly Atwood, Luis Buenaver, Michael Smith, Chien-Ming Huang. **Re-imagining Behavioral Sleep Medicine: Designing Conversational Sleep Diary and Visualization Tool.** *Under review*, 2025. [arXiv:2509.15378](https://arxiv.org/abs/2509.15378).
3. Amama Mahmood*, Bokyung Kim*, **Honghao Zhao**, Molly Atwood, Luis Buenaver, Michael Smith, Chien-Ming Huang. **Evaluating an LLM-Powered Voice Assistant for Sleep Diary Intake Through a Month-Long Deployment.** *In preparation*, 2025.

Teaching Experience

Boston University, Department of Computer Science

Teaching Assistant, CS460

Jan 2023 - Jul 2023

Skills

Programming: Python, Java, C/C++, SML, OCaml, Kotlin, Swift. **Databases:** SQL, NoSQL.

ML & Robotics: PyTorch, ROS, dVRK, Unity, SAM2, FoundationPose, Co-Tracker.

Languages: Mandarin (native), English (fluent). **Tools:** Microsoft Office, advanced Excel formulas.

Interests: LEGO, snowboarding, badminton, basketball.